



Grade 4 Science

OPTIMA ACADEMY ONLINE · FAMILY COURSE OVERVIEW · GRADE 4 · FULL YEAR

160 min per week · 2 lessons weekly

About this overview. This document is an example of how the course might be taught. More investigations and examples are approved for this course than appear here, so scholars may see some variation in activities and projects from course to course, and between the live school and On-Demand. Each teacher also leans into their own strengths and passions; the structure, expectations, and time commitment below are representative of the course as designed.

Grade 4 Science carries scholars from the night sky to the inside of a circuit, woven together by threads of exploration, map-making, resource use, and environmental stewardship. The year runs as 32 modules, each with a concept lesson and an applied lesson, aligned to both NGSS and Florida's CPALMS benchmarks. Scholars practice a steady evidence routine — observe, record, model, explain, revise — and grow from teacher-guided explanations early in the year to independent evidence-based arguments by the capstone.

THE YEAR AT A GLANCE

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| 1 Thinking Like a Scientist (Module 1)
Observation, evidence, scientific tools, and designing fair investigations. | 2 Earth and Sky (Modules 2–3)
Earth's rotation and seasons, the Moon and its phases, star patterns, and how space exploration affects Florida. |
| 3 Maps and Earth's Changing Surface (Modules 4–5)
Landform maps, landscapes changing over time, and weathering and erosion investigations. | 4 Rocks, Minerals, and Resources (Modules 6–8)
Rock types, mineral properties, and Florida's renewable and nonrenewable resources. |
| 5 Ecosystems and Living Things (Modules 9–15)
Energy from Sun to life, food webs, plant and animal structures, senses and the brain, inheritance, life cycles, and human impact. | 6 Natural Forces, Safety, and Science Tools (Modules 15–17)
Natural hazards, an engineering challenge to build for safety, and how microscopes and telescopes extend observation. |
| 7 Energy, Motion, and Collisions (Modules 18–24)
Energy transfer, collisions, speed, and designing a device that converts energy. | 8 Matter and Its Changes (Modules 25–28)
Properties of matter, magnets, and physical and chemical changes. |
| 9 Waves, Light, and Communication (Modules 29–30)
Wave models, how light lets us see, and designing a way to send messages with wave patterns. | 10 Energy Sources and the Environment (Module 31)
Comparing energy sources by benefits, risks, and environmental effects. |
| 11 Capstone: Think Like Scientists (Module 32)
Scholars apply the year's evidence, models, and explanations to a final inquiry. | |

A TYPICAL WEEK

- Two science lessons each week, completing one module.
- The first lesson introduces the week's core idea with readings, examples, and guided observation.
- The second applies it: an investigation, data or map analysis, model, or engineering design task with criteria, testing, and revision.

TIME COMMITMENT

160 minutes / week

two lessons a week, about 80 minutes each

- Each module spans one week: one concept lesson, one applied lesson.
- Engineering challenges and the capstone may run a little longer.

WHAT SCHOLARS NEED

- A science notebook for observations, evidence tables, and models.
- Simple household materials for investigations and design challenges.
- A computer with reliable internet and access to Canvas.

On-Demand (asynchronous). Scholars move through the course at their own pace, with due dates to keep the year on track. An Optima teacher is available throughout for support, feedback, and grading.

Live school. Live-school scholars take this same course on Optima's VR campus with a teacher and classmates, meeting twice a week on the school's weekly schedule.